

Martel Vasquez
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[Linkedin](#) | [GitHub](#) | [Personal Website](#)

EDUCATION

George Mason University, Fairfax, VA

Anticipated Graduation: August 2025

- Bachelor of Science in Computer Science, Minor in Software Engineering
- GPA: 3.75
- Dean's List: Spring 2024, Summer 2024

RELEVANT COURSEWORK

- Data Structures & Algorithms, Secure Programming and Systems, Formal Methods & Models, Computer Systems & Programming, Object-Oriented Programming, Essentials of CS, Calculus 1-3

SKILLS

- **Languages:** Java, Python, C, HTML, CSS
- **Tools/Technologies:** GitHub, Linux, OS/161, GitLab, VS Code, IntelliJ, CLion, JUnit

TECHNICAL PROJECTS

- **Java**
 - **Tetris Game:** Developed a Tetris game using a 2-D dynamic array to manage game elements efficiently, increasing game performance by 15%.
 - **Cache Simulator:** Created a cache simulator utilizing linked lists and HashMap to model FIFO, LFU, and LRU cache strategies, reducing cache retrieval times by 20%.
 - **Image Processor:** Built an image processor using the Quadtree data structure to process and manipulate PGB images, improving processing speed by 30%.
 - **Bellman-Ford Algorithm Simulator:** Developed a simulator to compute the shortest path in a weighted directed graph, showcasing proficiency in algorithm implementation and graph theory.
- **C**
 - **Otur CPU Scheduler:** Developed a CPU scheduler for the StrawHat Virtual Machine, managing process initialization, insertion, removal, and state transitions with three linked list queues. Improved scheduling efficiency by 30%.
 - **Shell and Buffer Management System:** Created an interactive command-line interpreter that emulated a shell, utilizing programming concepts such as forking child processes, executing programs, signal handling, file redirection, and pipes.
- **Python**
 - **Student Profile Management System:** Developed a system that implements functionalities for reading, creating, updating, and deleting student profiles using file I/O operations and dictionaries, collaborating with a utility module. Reduced data retrieval time by 15%.
- **HTML & CSS**
 - **Personal Website:** Designed and developed a personal website using HTML, CSS, and JavaScript, showcasing my projects and skills. This project enhanced my web development proficiency and improved my understanding of user experience design.

WORK EXPERIENCE

Footwear Associate, Dicks Sporting Goods, Fairfax, VA

August 2021 – Present

- Improved communication and selling skills, contributing to a 35% increase in sales.
- Delivered excellent customer service, helping boost positive store reviews by 20%.